

Partial Solutions to Practice Set 10

① $x(t) = e^{-t^2}$ for all $t \in \mathbb{R}$.

(a) ROC: $\sigma \geq \sigma_0$

$$\int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt = \int_{-\infty}^{+\infty} e^{-t^2} \cdot e^{-\sigma t} dt$$

$$= \int_{-\infty}^{+\infty} e^{-(t^2 + \sigma t)} dt = \int_{-\infty}^{+\infty} e^{-\left[\left(t + \sigma/2\right)^2 - \sigma^2/4\right]} dt$$

$$= e^{\sigma^2/4} \int_{-\infty}^{+\infty} e^{-(t + \sigma/2)^2} dt$$

$$= e^{\sigma^2/4} \cdot \sqrt{\pi} < +\infty.$$

$$\Rightarrow \int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt < +\infty \text{ for all } s \in \mathbb{C}.$$

$\Rightarrow$ ROC is the entire s -plane.

(b) $X(s)$:

$$X(s) = \int_{-\infty}^{+\infty} e^{-t^2} \cdot e^{-st} dt$$

$$= \int_{-\infty}^{+\infty} \exp\left[-(t^2 + st)\right] dt$$

$$= \int_{-\infty}^{+\infty} \exp \left[- \left\{ (t+s/2)^2 - s^2/4 \right\} \right] dt$$

$$= e^{s^2/4} \int_{-\infty}^{+\infty} \exp \left[- (t+s/2)^2 \right] dt$$

$$= \sqrt{\pi} \cdot e^{s^2/4}.$$

Sanity-check:

The ROC includes the $j\omega$ axis $\Rightarrow$ Fourier transform exists.

$$\text{Use } s=j\omega \text{ in } X(s) = \sqrt{\pi} \cdot e^{s^2/4}$$

$$\Rightarrow X(j\omega) = \sqrt{\pi} \cdot e^{-\omega^2/4}$$

which is indeed the Fourier transform of e^{-t^2} .

② (a) ROC of $x(t) = \frac{1}{1+t^2}$:

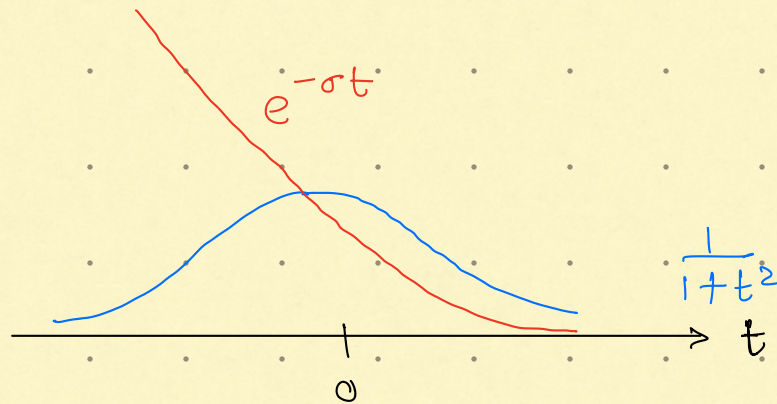
$$\text{Let } s = \sigma + j\omega$$

$$\int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt = \int_{-\infty}^{+\infty} \frac{1}{1+t^2} \cdot e^{-\sigma t} dt,$$

If $\sigma = 0$: $\int_{-\infty}^{+\infty} \frac{1}{1+t^2} e^{-\sigma t} dt = \int_{-\infty}^{+\infty} \frac{1}{1+t^2} dt < +\infty,$

$\Rightarrow$ The line $\text{Re}(s)=0$, i.e., the jw axis is in the ROC.

If $\sigma > 0$:



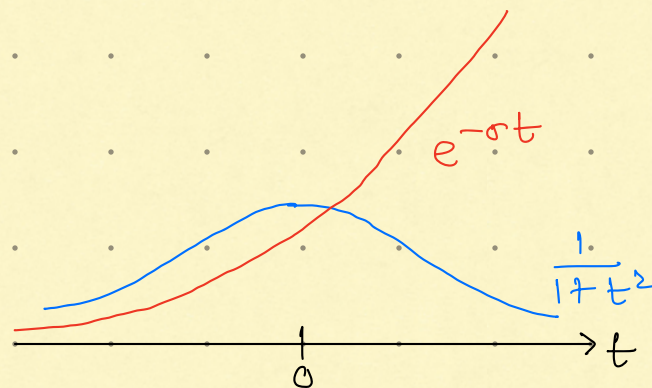
$$\int_{-\infty}^{+\infty} \frac{e^{-\sigma t}}{1+t^2} dt \geq \int_{-\infty}^0 \frac{e^{-\sigma t}}{1+t^2} dt$$

$$= \int_0^{\infty} \frac{e^{\sigma t}}{1+t^2} dt$$

$$= +\infty \quad \text{since } \frac{e^{\sigma t}}{1+t^2} \rightarrow +\infty \text{ as } t \rightarrow +\infty.$$

$\Rightarrow$ Any s with $\text{Re}(s) > 0$ is NOT in ROC

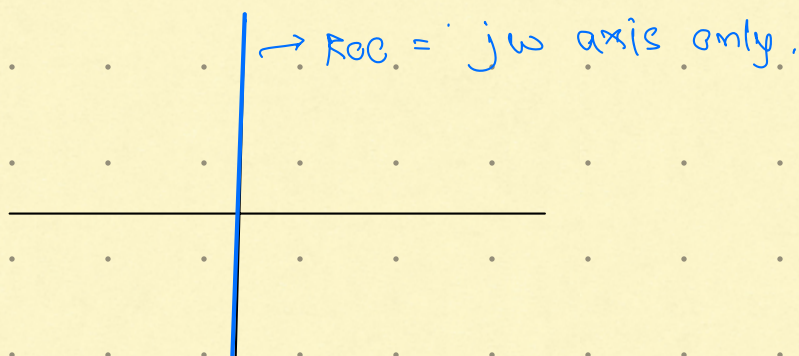
If $\sigma < 0$



$$\int_{-\infty}^{+\infty} e^{-\sigma t} \frac{1}{1+t^2} dt \geq \int_0^{\infty} \frac{e^{-\sigma t}}{1+t^2} dt = +\infty \quad \text{since}$$

$$\frac{e^{-\sigma t}}{1+t^2} \rightarrow +\infty \text{ as } t \rightarrow +\infty.$$

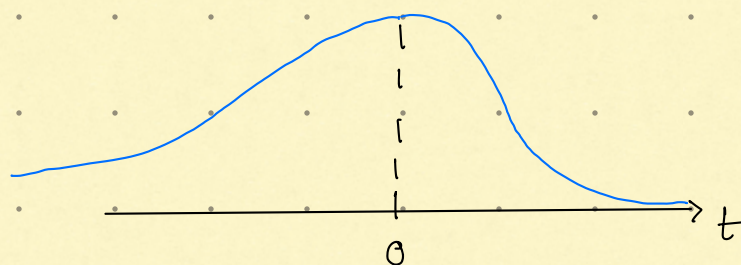
To conclude: $\text{ROC} = \{s: \text{Re}(s) = 0\}$
 $= j\omega \text{ axis}$



(b) $x(t) = \begin{cases} e^{-t^2} & \text{if } t > 0 \\ \frac{1}{1+t^2} & \text{if } t < 0 \end{cases}$

$\rightarrow$ decays very fast as $t \rightarrow \infty$.

$\rightarrow$ decays slowly as $t \rightarrow -\infty$



Let $s = \sigma + j\omega$.

$$\int_{-\infty}^{+\infty} |x(t)| e^{-\sigma t} dt = \underbrace{\int_{-\infty}^0 \frac{e^{-\sigma t}}{1+t^2} dt}_{I_1} + \underbrace{\int_0^{\infty} e^{-t^2} e^{-\sigma t} dt}_{I_2}$$

$$I_1 = \int_0^{\infty} \frac{e^{\sigma t}}{1+t^2} dt$$

$\rightarrow$ If $\sigma > 0$: $I_1 = +\infty$.

$\rightarrow$ If $\sigma \leq 0$: $I_1 < +\infty$

$$\text{If } \sigma > 0: \frac{e^{\sigma t}}{1+t^2} \rightarrow +\infty \text{ as } t \rightarrow +\infty$$

$$\Rightarrow I_1 = +\infty$$

$$\text{If } \sigma = 0 \text{ or } \sigma < 0: I_1 \leq \int_0^{\infty} \frac{1}{1+t^2} dt \text{ since } e^{\sigma t} \leq 1$$

$$\leq +\infty$$

$$I_2 = \int_0^{\infty} e^{-t^2} \cdot e^{-\sigma t} dt \leq \int_{-\infty}^{+\infty} e^{-t^2} \cdot e^{-\sigma t} dt$$

From question ① we know $\int_{-\infty}^{+\infty} e^{-t^2} \cdot e^{-\sigma t} dt < +\infty$ for

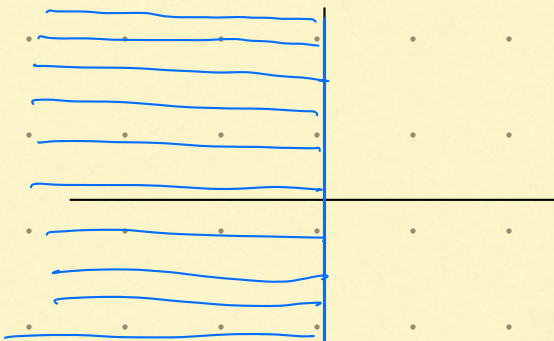
any σ .

$$\Rightarrow I_2 < +\infty \text{ for any } s \in \mathbb{C}.$$

Notice that $\int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt$ is finite if & only if

both I_1 and I_2 are finite:

$$\Rightarrow \text{ROC} = \{s: \operatorname{Re}(s) \leq 0\}.$$



The boundary line

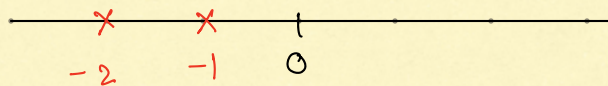
$$\operatorname{Re}(s) = 0$$

belongs to the ROC.

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$$X(s) = \frac{1}{(s+1)(s+2)}$$

$$\text{Poles} = -1, -2.$$



(a) Three possible ROCs: (strict inequalities)

$$(i) \operatorname{Re}(s) < -2$$

$$(ii) -2 < \operatorname{Re}(s) < -1$$

$$(iii) \operatorname{Re}(s) > -1.$$

(b) Solution for ROC (ii) above:

$$\begin{aligned} X(s) &= \frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2} \\ &= \frac{(A+B)s + (2A+B)}{(s+1)(s+2)} \end{aligned}$$

$$\begin{aligned} 2A+B &= 1 \\ A+B &= 0 \end{aligned} \Rightarrow A=1, B=-1$$

$$\therefore X(s) = \frac{1}{s+1} + \frac{(-1)}{s+2}, \quad -2 < \operatorname{Re}(s) < -1$$

Now, invert term by term by using position

of ROC with respect to each pole.

$\frac{1}{s+1}$: ROC is to the left of pole $s = -1$.

$\Rightarrow$ left-sided signal.

(Recall: $-e^{-at}u(-t) \leftrightarrow \frac{1}{s+a}$, $\text{Re}(s) < -\text{Re}(a)$.)

$$\therefore \rightarrow -e^{-t}u(-t).$$

$\frac{-1}{s+2}$: ROC is to the right of pole $s = -2$.

$\Rightarrow$ right-sided signal

$$\rightarrow -e^{-2t}u(t).$$

$$x(t) = -e^{-2t}u(t) - e^{-t}u(-t).$$

4.

$$\begin{aligned} \textcircled{a} \quad \sin(\omega_0 t + \phi_0) &= \text{Im} \{ e^{j\omega_0 t} e^{j\phi_0} \} \\ &= \frac{e^{j\phi_0} e^{j\omega_0 t} - e^{-j\phi_0} e^{-j\omega_0 t}}{2j} \end{aligned}$$

$$e^{j\omega_0 t} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s - j\omega_0}, \quad \text{Re}(s) > 0$$

$$e^{-j\omega_0 t} u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s + j\omega_0}, \quad \text{Re}(s) > 0$$

$$\sin(\omega_0 t + \phi_0) u(t) \xleftrightarrow{L} \frac{1}{2j} \left[\frac{e^{j\phi_0}}{s - j\omega_0} - \frac{e^{-j\phi_0}}{s + j\omega_0} \right],$$

$$\operatorname{Re}(s) > 0$$

$$= \frac{e^{j\phi_0} (s + j\omega_0) - e^{-j\phi_0} (s - j\omega_0)}{2j \times (s^2 + \omega_0^2)}, \quad \operatorname{Re}(s) > 0,$$

$$(b) \quad e^{-a|t|} = e^{-at} u(t) + e^{at} u(-t).$$

$$(i) \text{ ROC: } \int_{-\infty}^{+\infty} e^{-a|t|} e^{-\sigma t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-\sigma t} dt + \int_{-\infty}^{+\infty} e^{at} u(-t) e^{-\sigma t} dt$$

Both integrals on the RHS must be finite.

$$\Rightarrow \text{ROC} = \text{ROC of } e^{-at} u(t) \rightarrow \operatorname{Re}(s) > -a$$

intersection

$$\text{ROC of } e^{at} u(-t) \rightarrow \operatorname{Re}(s) < a$$

$$\therefore \text{ROC} = -a < \operatorname{Re}(s) < +a$$

$$(ii) \quad X(s) = \text{Laplace of } e^{-at} u(t) + \text{Laplace of } e^{at} u(-t)$$

$$= \frac{1}{s+a} - \frac{1}{s-a}, \quad -a < \operatorname{Re}(s) < +a$$

$$= \frac{-2a}{s^2 - a^2}, \quad -a < \operatorname{Re}(s) < +a.$$

⑤ LTI system is stable $\Rightarrow \int_{-\infty}^{+\infty} |h(t)| dt < +\infty.$

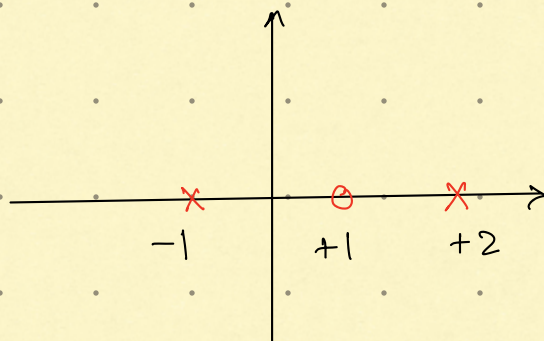
Check if $j\omega$ axis is in ROC

i.e. $\operatorname{Re}(s) = 0.$

$$\begin{aligned} \int_{-\infty}^{+\infty} |h(t) \cdot e^{-st}| dt &= \int_{-\infty}^{+\infty} |h(t)| \cdot e^{-\sigma t} dt, \quad \sigma = 0 \\ &= \int_{-\infty}^{+\infty} |h(t)| dt < +\infty. \end{aligned}$$

Yes, $j\omega$ axis belongs to ROC.

⑥ Pole-Zero Plot:



Three possibilities of ROC:

$$\operatorname{Re}(s) < -1$$

$-1 < \operatorname{Re}(s) < +2 \rightarrow$ only this includes the $j\omega$ axis
(system is stable)

$$\operatorname{Re}(s) > +2$$

$\Rightarrow$ ROC is $-1 < \text{Re}(s) < +2$.

(d) Since ROC is not a right-half plane and not a left-half plane:

$h(t)$ is a two-sided signal.

$\Rightarrow h(t)$ can not be a causal signal.

$\Rightarrow$ LTI system is non-causal.

(e) If LTI system is causal:

$\Rightarrow h(t)$ is a right-sided signal

$\Rightarrow$ ROC is to the right of rightmost pole

$\Rightarrow \text{ROC} = \text{Re}(s) > +2$.

Use the ROC to do inverse Laplace transform via partial fractions.

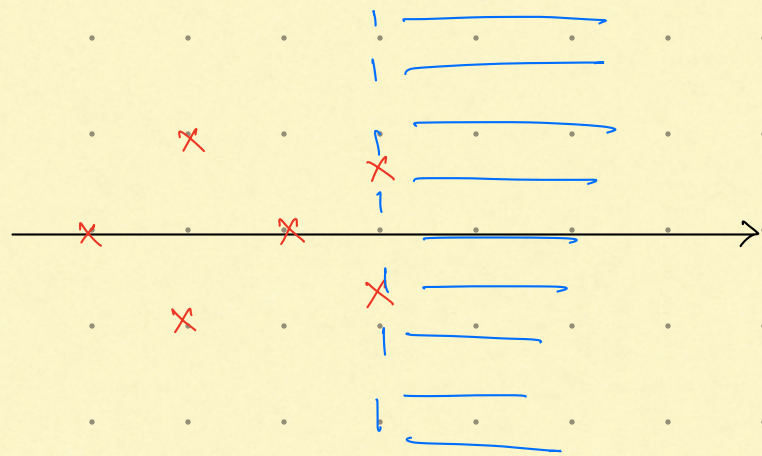
(6) To prove:

System Stable
& Causal

$\Rightarrow$

all poles have -ve
real parts.

System is causal $\Rightarrow$ ROC is to the right of right most pole



System is stable: $j\omega$ axis is inside ROC
i.e. in the blue region above

$\therefore$ all poles have negative real parts.

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$$(a) \quad y'' + \frac{R}{L} y' + \frac{1}{LC} y = \frac{1}{LC} x$$

$$\Rightarrow s^2 Y(s) + \frac{R}{L} s Y(s) + \frac{1}{LC} Y(s) = \frac{1}{LC} X(s)$$

$$\Rightarrow \frac{Y(s)}{X(s)} = \frac{1/LC}{s^2 + s \cdot \frac{R}{L} + \frac{1}{LC}}$$

$$H(s) = \frac{1/LC}{s^2 + \frac{R}{L} s + \frac{1}{LC}}$$

$$(b) \quad \frac{1}{Lc} = \omega_0^2 \quad 2\epsilon = R\sqrt{\frac{c}{L}}$$

$$2\epsilon \cdot \omega_0 = R \sqrt{\frac{c}{L}} \times \frac{1}{\sqrt{L}} \times \frac{1}{\sqrt{c}} = \frac{R}{L}$$

$$\therefore H(s) = \frac{\omega_0^2}{s^2 + 2\epsilon\omega_0 s + \omega_0^2}$$

(c) Two poles: roots of $s^2 + 2\epsilon\omega_0 s + \omega_0^2$

$$= \frac{-2\epsilon\omega_0 \pm \sqrt{4\epsilon^2\omega_0^2 - 4\omega_0^2}}{2}$$

$$= \frac{-2\epsilon\omega_0 \pm 2\omega_0\sqrt{\epsilon^2 - 1}}{2}$$

$$= -\epsilon\omega_0 \pm \omega_0\sqrt{\epsilon^2 - 1}$$

(d) (i) $\epsilon > 0$: $\epsilon^2 - 1 > 0$

$$\therefore \omega_0\sqrt{\epsilon^2 - 1} > 0$$

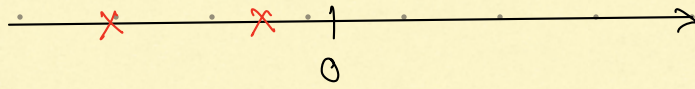
$$\therefore \text{poles are } -\epsilon\omega_0 - \omega_0\sqrt{\epsilon^2 - 1}, \rightarrow \text{clearly -ve}$$

$$-\epsilon\omega_0 + \omega_0\sqrt{\epsilon^2 - 1}$$

For the second pole:

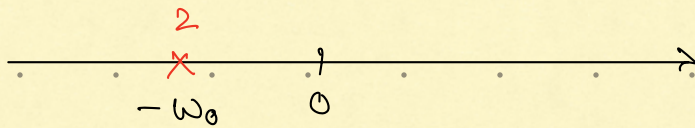
$$\text{observe that } \sqrt{\epsilon^2 - 1} < \epsilon \Rightarrow \omega_0\sqrt{\epsilon^2 - 1} < \epsilon\omega_0$$

$$\therefore -\epsilon \omega_0 + \omega_0 \sqrt{\epsilon^2 - 1} < 0,$$



(ii) $\epsilon = 1$;

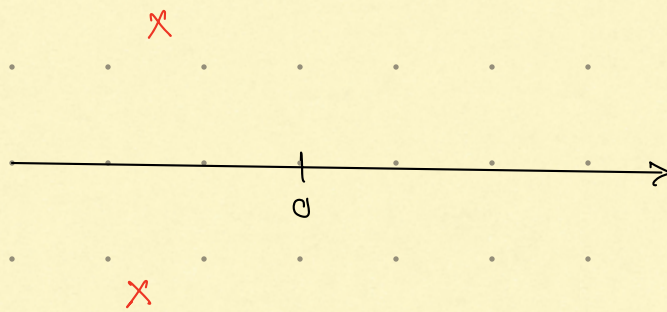
Two poles at $-\omega_0$,



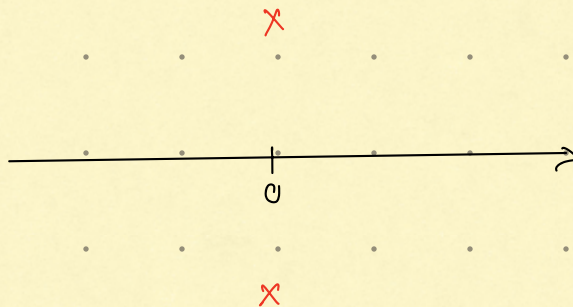
(iii) $\epsilon < 1$;

Poles: $-\epsilon \omega_0 \pm j \omega_0 \sqrt{1 - \epsilon^2}$

$$= -\epsilon \omega_0 \pm j \omega_d$$



(iv) $\epsilon = 0$: Poles at $\pm j \omega_0$



(c) Solution for $0 < \epsilon < 1$:

Poles at: $-\epsilon\omega_0 \pm j\omega_d$, $\omega_d = \omega_0 \sqrt{1-\epsilon^2}$.

$$H(s) = \frac{\omega_0^2}{s^2 + 2\epsilon\omega_0 s + \omega_0^2}$$

$$= \frac{\omega_0^2}{(s + \epsilon\omega_0)^2 + \omega_0^2 - \epsilon^2\omega_0^2}$$

$$= \frac{\omega_0^2}{(s + \epsilon\omega_0)^2 + \omega_d^2}$$

$$= \frac{\omega_0^2}{\omega_d} \times \frac{\omega_d}{(s + \epsilon\omega_0)^2 + \omega_d^2}$$

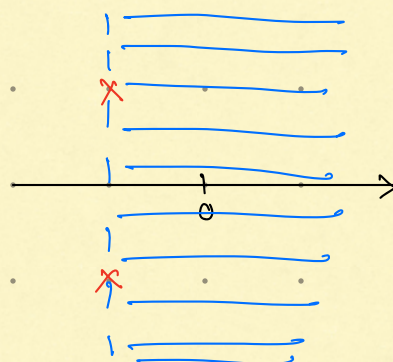
LTI system is causal

$$\Rightarrow h(t) = 0 \quad \forall t < 0$$

$\Rightarrow h(t)$ is right-sided

$\Rightarrow$ ROC of $H(s)$ is to the right of the right-most pole

$$\Rightarrow \text{ROC: } \operatorname{Re}(s) > -\epsilon\omega_0$$



To find inverse Laplace of

$$\frac{\omega_0^2}{\omega_d} \times \frac{\omega_d}{(s + \epsilon \omega_0)^2 + \omega_d^2}, \quad \text{Re}(s) > -\epsilon \omega_0$$

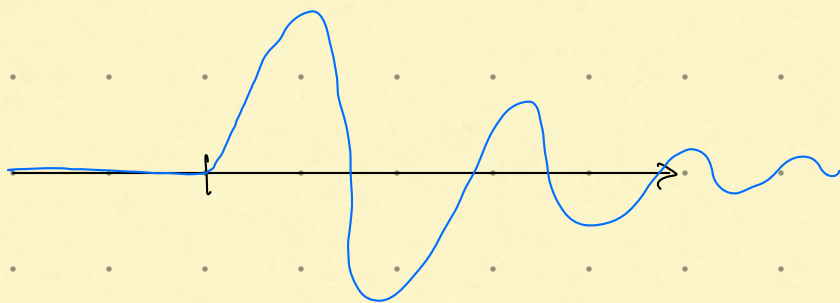
Remember the formula:

$$e^{-at} \sin \omega_d t \, u(t) \xleftrightarrow{L} \frac{\omega_d}{(s+a)^2 + \omega_d^2}, \quad \text{Re}(s) > -a$$

($a \in \mathbb{R}$)

$$\therefore h(t) = \frac{\omega_0^2}{\omega_d} \cdot \underbrace{e^{-\epsilon \omega_0 t}}_{\substack{\downarrow \\ \text{exponential} \\ \text{decay}}} \cdot \underbrace{\sin(\omega_d t) \cdot u(t)}_{\substack{\downarrow \\ \text{sine wave from} \\ t=0 \text{ onwards.}}}$$

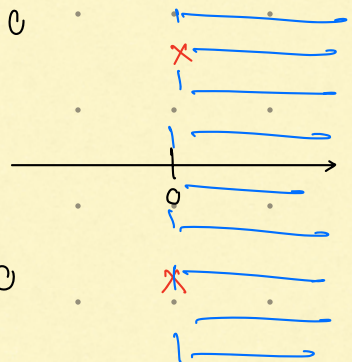
↓
Reduces amplitude of sine wave to 0
as $t \rightarrow +\infty$.



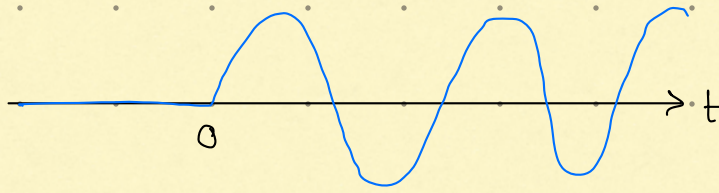
$$(f) \quad H(s) = \omega_0 \times \frac{\omega_0}{s^2 + \omega_0^2}, \quad \text{Re}(s) > 0$$

Use formula:

$$\sin \omega_0 t \, u(t) \xleftrightarrow{L} \frac{\omega_0}{s^2 + \omega_0^2}, \quad \text{Re}(s) > 0$$

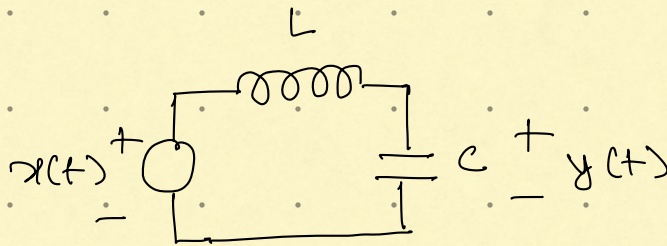


$$\Rightarrow h(t) = \omega_0 \sin(\omega_0 t) u(t)$$



The amplitude of sine wave is not damped as $t \rightarrow +\infty$.

This happens when $\epsilon = 0$ i.e. $R = 0$.



This is an ideal resonant circuit excited with an impulse at $t=0$.

The resulting voltage $y(t)$ continues to oscillate at frequency $\omega_0 = \frac{1}{\sqrt{LC}}$ for all time $t > 0$.